

Welcome to the amos2025ss04-ai-driven-testing wiki!

Our project

How to use

[User and Developer Guide](#) For a quick start and for the correct setup. You may also want to check out specific sections for certain applications:

- [How to start the Webinterface Frontend](#) Tutorial on how to start it
- [Ollama Setup Information](#) Possible ways to use Ollama to run LLMs
- [Possible Ways to run Ollama and Settings](#)

How to work on this project

- [Contributing](#)
- [How to be a Release Manager](#) How to do the weekly release. This is relevant or any future team tat takes part in the AMOS course.

Structure

- [Architecture](#) Overview of the system architecture and component interactions.
- [Backend API Design](#) The structure recieving prompts, interacting with LLMs and returning the results to the client.
- [CI Pipeline](#) CI/CD Pipeline Overview over the continuous integration pipeline for the project.
- [Execution Flow Design](#) TODO: finish this entry
- [Model Configuration Design](#) How we handel models.
- [Module System Design](#) How we handel the extentions.
- [Prompt Design](#) Some Guidance on effective prompting.
- [Robot Framework](#) Information about potential integration with the Robot Framework.

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Models

A list of all language models evaluated for use in the project. We only chose the versions of models that are available under an open source license.

- [DeepCoder](#)
- [DeepSeek-Coder V1](#)
- [Google Gemma 3](#)
- [LLMs incompatibility with our project](#) Overview of models that were evaluated but deemed unsuitable.
- [Mistral AI](#)
- [OpenHermes 2.5](#)
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- [Phi-4 Reasoning](#)
- [Qwen 2.5 Coder](#)
- [Qwen3](#)
- [Smollm2](#)
- [StarCoder](#)
- [StarCoder2](#)
- [TinyLlama](#)

Our research and experiments

- [Docker Performance](#) Performance of different docker configurations when running LLMs
- [Evaluating Large Language Model Responses to Spelling Errors](#)
- [LLM Code Understanding Evaluation](#)
- [Comparison of 1b, 3b, 7b and 14b models](#)
- [Running AI LLM Projects in CI](#)

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<https://github.com/amosproj/amos2>



LLM components and assessment tools

- [AI-Model-Benchmark](#) Standard Benchmarks used to evaluate LLMs
- [Benefits of chaining LLMs](#)
- [Code Complexity](#) Description of the code complexity metrics that will be used for evaluation, including MCC and CCC.
- [Code Coverage](#)
- [Include a project as context](#) This explains how to include repositories into the LLM prompt as context using a RAG and the alternatives.
- [Iterative Refinement \(Multi-Pass Generation\)](#) Passing the response as the new prompt input.

Training an LLM by yourself

- [How to train a LLM](#)
- [How to run on the FAU HPC](#)
- [Choosing the Right Dataset for LLM Training on the University HPC](#)
- [How to finetune a LLM](#)

The maintenance of this Wiki officially stops at 16.07.2025, as this is the Demo-day of our project and no further contribution is expected from the team.